

Tao 2019 — Almost all orbits of the Collatz map attain almost bounded values

PDF: tao-2019-almost-all-orbits.pdf (symlink → ~/src/collatz-cryptid/data/refs/ ; gitignored — copyrighted, this repo may eventually go public)

Source: arXiv:1909.03562 (v5, 2022-02-15), <https://arxiv.org/abs/1909.03562>

Published: Terence Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi **10** (2022), e12. DOI: 10.1017/fmp.2022.8

Length: 57 pp + figures. MSC 37P99.

What it is

The strongest known "almost all" result on the Collatz conjecture. For ANY function $f : \mathbb{N}+1 \rightarrow \mathbb{R}$ with $f(N) \rightarrow \infty$ — however slowly, e.g. $\log \log \log \log N$ — the Collatz orbit minimum satisfies $\text{Col_min}(N) < f(N)$ for almost all N , in **logarithmic density** (Theorem 1.3). Supersedes Terras/Everett ($< N$), Allouche ($< N^\theta$, $\theta > 0.869$), and Korec ($\theta > \log 3 / \log 4 \approx 0.7924$), all of which used natural density and fixed powers.

This repo formalizes exactly this paper; the node-by-node decomposition lives in `./BLUEPRINT.md` (26 nodes, design decisions D1–D8) — this summary is the paper-facing companion, not a duplicate of the ledger.

Proof architecture (one breath)

Work with the Syracuse map $\text{Syr}(N) = (3N+1)/2^{v_2(3N+1)}$ on odd numbers. The n -fold iterate is an explicit affine map determined by the valuation vector

$\vec{a}(N) = (v_2(3N+1), v_2(3\text{Syr}(N)+1), \dots)$ (eq. 1.7). For N drawn log-uniformly, $\vec{a}(N)$ is close in total variation to iid $\text{Geom}(2)$ (Prop 1.9, §4) — so the 3-adic residue $\text{Syr}^n(N) \bmod 3^n$ is governed by the **Syracuse random variables**

$\text{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) = F_n(\text{Geom}(2)^n) \bmod 3^n$ (1.21). The heart of the paper: these become equidistributed at fine 3-adic scales (Prop 1.14), which follows from superpolynomial decay of their characteristic function, $\mathbb{E} e(-\xi \cdot \text{Syrac}/3^n) \ll_A n^{-A}$ uniformly in $\xi \not\equiv 0 \pmod 3$ (Prop 1.17, §7 — the hard section). Fine-scale mixing feeds a Bourgain-style invariant-measure surrogate: the first-passage locations $\text{Pass}_x(N_y)$ stabilize in distribution across scales $y = x^\alpha$ vs $x^{\{\alpha^2\}}$ (Prop 1.11, §5), and iterating the passage map across a telescoping tower of scales pulls almost every orbit below any slowly-growing f (Thm 3.1 → 1.6 → 1.3, §3).

§7 proves Prop 1.17 by pairing the geometric variables ($b_j = a_{\{2j-1\}} + a_{\{2j\}} \sim \text{Pascal}$), factoring the characteristic function into an average of cosines over a **2-D renewal**

process, and showing the walk keeps hitting "white" points (where the cosine contracts). The black set — where no contraction happens — is exactly a union of well-separated **triangles** in (j, l) -space (Lemma 7.4, elementary 3-adic number theory), and the walk's drift (slope 4 vs triangle-diagonal slope $\log 9 / \log 2 \approx 3.17$) forces it out of any triangle, through the white buffer Lemma 7.4 guarantees around it (Prop 7.8's three-case induction, Lemmas 7.9/7.10).

Key recallable facts

- **Log density, not natural density:** forced by the $\exp(O(\sqrt{n}))$ multiplicative spread of $\text{Syr}^n(N) \approx (3/4)^n N$ (1.17); log-uniform measure is what the Syracuse flow approximately transports (Remark 1.16 sketches the natural-density upgrade as plausible but unpursued).
- **Quantitative core** (Thm 3.1): $P(\text{Col_min}(\text{Log}(\mathbb{N}[1, x])) \leq N_\theta) \geq 1 - O(\log^{-c} N_\theta)$ uniformly in $x \geq 2$; Remark 1.4: sharpening to a constant C_θ would be nearly Collatz-complete (heuristic obstruction), so this "almost bounded" form is the natural wall.
- **The paper's own waypoint** (Remark 5.1): the §4/§5 machinery alone already recovers $\text{Syr_min}(N) \leq N^\theta$ for a.a. N (Korec-strength) — our campaign's Series- β milestone.
- **Syrac($\mathbb{Z}/9\mathbb{Z}$) table** (p.10, our harness check 3): values 0..8 have masses $0, 8/63, 16/63, 0, 11/63, 4/63, 0, 2/63, 22/63$ — multiples of 3 are never hit.
- **Footnote 6 trap:** §7 REVERSES the variable order of (1.5) $\rightarrow$ (1.26); same law since $\text{Geom}(2)^n$ is exchangeable. Encoded in our skeleton as `syracZ_eq_rev_fnat`.
- **Only two geometric distributions appear:** $\text{Geom}(2)$ (valuations) and $\text{Geom}(4)$ (renewal holding-time first coordinate); $\text{Pascal} = \text{Geom}(2) + \text{Geom}(2)$; Hold has mean (4,16).
- Constants: $\alpha = 1.001$ (1.18); §7's $\varepsilon < 1/100$ absolute; A -dependence threads $C_A \rightarrow C_{\{A, \varepsilon\}} \rightarrow P \rightarrow R = \lfloor A^2 / \varepsilon^4 \rfloor$ (our D7 ledger).
- Acknowledgments note the 2-D renewal formulation is due to a suggestion of **Marek Biskup**;
Ben Green suggested the diagonalisation remark (fn. 2).

Why it matters to the Lean work

- Greenfield: no formalization exists in any prover (checked 2026-07-08); this repo is the first attempt. Everything in the paper is *discrete* probability — hence our D1 (PMF/tsum, no measure theory) and D6 (finitized renewal recursion) design bets, both kernel-validated.
- The three formalization risk kernels map to: Lemma 2.2 (local 2-D Gaussian bounds; the paper's one complex-analytic proof, we take a real-variable tilting route — D5), Lemma 7.4 (triangle structure), and §7.4's Cases 2–3 (Lemmas 7.9/7.10). See `../EXECUTABILITY.md`.

- Related reading in the sibling repo: Lagarias surveys + Krasikov–Lagarias density bounds (`~/src/collatz-cryptid/data/refs/` , indexed at `collatz-cryptid/notes/refs.md`).